

PROJECT ABSTRACT SUBMISSION

COURSE: ADAPTIVE SOFTWARE ENGINEERING (24CI3201)

A.Y. 2026 - 2027

TITLE OF THE PROJECT	AI-Powered Automated Release Notes and Changelog Generation from Commit History	
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Project Abstract

Software development teams make frequent code changes and maintain detailed commit histories using version control systems such as Git. Preparing release notes and changelogs manually from these commits is a time-consuming process and may lead to incomplete or inconsistent documentation. This project proposes an **AI-Powered Automated Release Notes and Changelog Generation System** that analyzes commit history and automatically generates structured, human-readable release notes.

The system collects commit messages from Git repositories using the GitHub API and categorizes them into features, bug fixes, enhancements, and maintenance updates. An AI model is then used to summarize the changes and generate professional release notes suitable for developers, testers, and project stakeholders. The generated output improves project documentation, enhances communication among team members, and reduces manual effort.

The project follows Software Engineering and DevOps practices by integrating version control, automated workflows, and continuous documentation generation. Technologies such as Python, FastAPI, React.js, GitHub API, Docker, and GitHub Actions will be used for implementation. The proposed system aims to improve software release management, increase development productivity, and provide accurate project change tracking in modern software development environments.